

UrbanOS

AI Operating System for Smarter Development

Product Requirement Document

Vision

UrbanOS is an AI-powered planning and development operating system that helps planners, architects, engineers, developers, governments, and landowners design, evaluate, optimize, and execute development projects of any scale.

The platform transforms a project brief into intelligent development options, generating optimized layouts, feasibility reports, regulatory compliance checks, construction plans, manpower requirements, and implementation roadmaps.

UrbanOS aims to become the operating system for planning and development, enabling faster, smarter, more sustainable, and economically efficient projects.

Problem Statement

Development projects currently require coordination between multiple professionals:

- Urban planners
- Architects
- Civil engineers
- Structural engineers
- Environmental consultants
- Quantity surveyors
- Government authorities

This process is:

- Slow
- Expensive
- Fragmented
- Prone to design errors

- Difficult to optimize
- Difficult to evaluate financially

Developers often miss opportunities to maximize land value, sustainability, efficiency, and regulatory compliance.

Proposed Solution

UrbanOS uses AI to generate and evaluate development options automatically.

Users upload project information such as:

- Site location
- Plot boundaries
- Project goals
- Budget
- Development type

The system then produces:

Design Alternatives

- 2-3 optimized masterplan options
- 2D layouts
- 3D concepts
- Visual walkthroughs

Compliance Engine

Automatic checking against:

- Building bylaws
- Development regulations
- Zoning rules
- Government policies
- Fire safety norms

- Environmental requirements
- Infrastructure standards

Feasibility Engine

Generates:

- Cost estimates
- Revenue estimates
- ROI calculations
- FAR/FSI utilization
- Land value optimization
- Risk analysis

Construction Intelligence

Generates:

- Construction phases
- Project timelines
- Resource schedules
- Material requirements
- Manpower requirements
- Contractor scopes

Sustainability Engine

Optimizes for:

- Walkability
- Mobility
- Energy efficiency
- Water management
- Green infrastructure
- Climate resilience

Target Users

Phase 1

- Urban planners
- Architects
- Real estate developers
- Construction companies
- Planning consultancies

Phase 2

- Government agencies
- Infrastructure authorities
- Industrial developers
- Smart city agencies

Phase 3

- Individual homeowners
- Landowners
- Investors
- Townships and city-scale developers

Example Inputs

House Project

- Plot size
- Budget
- Family requirements
- Location

Output:

- Multiple floor plans
- 3D views
- Cost estimate
- Construction roadmap

Township Project

- Site boundaries
- Population targets
- Development goals

Output:

- Masterplan options
- Road network
- Land-use allocation
- Utility infrastructure
- Financial feasibility

Infrastructure Project

- Corridor alignment
- Budget
- Design standards

Output:

- Alignment options
- Construction sequencing
- Cost estimates
- Resource planning

MVP Scope

Module 1: Project Intake

User enters:

- Site details
- Budget
- Project objectives

Module 2: AI Planning Assistant

Generates:

- Land-use recommendations
- Development concepts
- Initial layouts

Module 3: Compliance Checker

Checks:

- Building rules
- Zoning restrictions
- Regulatory requirements

Module 4: Feasibility Engine

Generates:

- Cost estimates
- ROI estimates
- Project viability score

Module 5: Report Generator

Creates:

- PDF reports
- Development summaries
- Compliance reports

Long-Term Vision

UrbanOS becomes the AI Operating System for Development.

A user should be able to enter:

"I own 20 acres in Jaipur and want a mixed-use development with a ₹300 crore budget."

UrbanOS should automatically generate:

- Development concepts
- Site layouts
- 3D visualizations
- Regulatory compliance checks
- Financial feasibility
- Construction plans
- Resource planning
- Implementation roadmap

within minutes instead of months.

Core Promise

UrbanOS helps organizations plan smarter developments by combining AI, planning intelligence, regulatory knowledge, financial feasibility, and construction execution into a single operating system.